Finding and Repairing Point Defects in a Model Glass with a Denoising Network

Report on the two-dimensional study

Stacy Cheruto Nicholaus Ladislaus Abubakar Chilala
Gentille Uwera Michel Sangwa Boniface Godwin[†]
College of Engineering, Carnegie Mellon University Africa
{ccheruto, eladisla, achilala, guwera, sbmichel, bgodwin}@andrew.cmu.edu
[†]Teaching assistant supervising this project

Abstract

Solids are never perfect. Here and there an atom is missing, an extra atom is squeezed in, or two atoms sit in each other's places. Such *point defects* are easy to spot in a crystal, where the atoms form a regular pattern and anything out of pattern stands out. A glass has no pattern: its atoms are frozen in disorder, and once a defect is made, the neighbouring atoms shift and largely hide it. We ask whether a neural network can find the defect from the positions and kinds of the atoms alone, and whether the glass can then be returned to the defect-free structure it came from. We work with a simulated two-dimensional glass, make the four standard point defects ourselves (vacancy, interstitial, Frenkel pair and antisite) and let the glass settle, so the right answer is always known. A graph neural network that is first trained to remove random noise from glasses, and then to label atoms, finds 69.5% of vacancies, 50% of interstitials, 49% of antisites and 45.3% of Frenkel pairs. It does not need the formula for the forces between atoms (the interatomic potential). Three measures that involve no learning find fewer: the energy of each atom, which does need the potential (28–32.7%), the number of unlike neighbours of each atom (27.3–31.5%) and the room each atom has (15–47.5%). Correcting the number or kind of atoms where the network points, and then letting the atoms settle, restores 53.5%, 38%, 30% and 12.7% of the original glasses. A second network, trained to undo the defect directly, adds nothing to this. In short training runs, starting from the noise-removal network instead of from random weights nearly doubles the defects found. Finding is what limits the result: when the true location is given, settling alone restores 42–80%.

1 Introduction

Every solid has faults in the arrangement of its atoms. The simplest are *point defects*, faults the size of one atom (Figure 1). A *vacancy* is a missing atom. An *interstitial* is an extra atom squeezed in between the others. A *Frenkel pair* is an atom knocked out of its place into a gap nearby, which leaves a vacancy and an interstitial together; it is the basic damage that radiation does. An *antisite* is a pair of atoms of different kinds that have exchanged places [10, 19]. Point defects matter far beyond their size: they decide how atoms move through a material, how it ages and how it stands up to radiation.

In a *crystal* the atoms sit on a regular repeating pattern, the lattice, like tiles on a floor. A defect is then easy to define and to find: compare the structure with the perfect pattern and see what is missing, extra or misplaced.

A *glass* has no such pattern [21]. Its atoms are packed as closely as in a crystal but without order, like a liquid frozen in place. A point defect can still be *made* in a glass, by removing, adding, moving or swapping an atom. But the atoms around it then shift until the forces on them balance again (the glass *relaxes*), and the fault is spread over a few tens of atoms. Early simulations found that most vacancies in a simple model glass do not survive relaxation as open holes [1], although some remain and behave like vacancies in a crystal [6]. This makes two questions hard:

Table 1: Terms used in this report.

Term	Meaning
Crystal, lattice	A solid whose atoms sit on a regular repeating pattern (the lattice).
Glass	A solid whose atoms are frozen in disorder, with no repeating pattern.
Species	The kind of atom. Our glass has two: large A atoms and small B atoms.
Point defect	A fault the size of one atom: an atom missing, extra, displaced or exchanged.
Site	The place where an atom sits.
Interatomic potential	The formula for the energy of two atoms at a given distance. It gives the forces between the atoms.
Relaxation (energy minimisation)	Letting the atoms shift until the forces on them balance. The structure then rests in a "valley" of the energy, a local minimum.
Periodic box	The simulated square of glass repeats in every direction: an atom that leaves on one side comes back on the other. This imitates a large sample without edges.
Cell of an atom	The patch of space that is closer to this atom than to any other (its Voronoi cell): the room the atom has.
σ, ε	Our units. σ is about one atom diameter; ε is the strength of the attraction between two A atoms.
Denoiser	A neural network trained to undo random shifts ("noise") of the atoms.

1. **Find.** Given a relaxed glass that contains one point defect, which atoms are wrong?
2. **Repair.** What is the defect-free glass it came from?

The usual tools are of two kinds. *Geometric* measures, such as the empty space around each atom [15], need only the positions of the atoms. But they are weak, because a glass is full of loosely and tightly packed places that are not defects. *Energy* measures are sharper, but they need the *interatomic potential*, the formula that gives the forces between atoms. For many materials no trusted potential exists, and structures seen under a microscope at atomic resolution, such as two-dimensional silica glass [9], come as positions only. A neural network that has learned what normal glass looks like could find what is abnormal from the structure alone. This is where a learned model can add something that a simulation does not already give. A simulated glass is the right place to test the idea, because there the defect and the original are known exactly.

This report gives the complete study in a two-dimensional model glass. It replaces the approach of our midterm report, which trained a model to *place* defects that we had defined ourselves. After a discussion with Prof. Ismaila Dabo we changed the defects to the standard ones, changed the task from placing defects to finding and repairing them, and reduced the method to one network whose every part has a stated reason. Table 1 explains the terms we use. Our findings are:

- A neural network, the *finder*, locates each of the four defects more often than three measures that involve no learning: the energy of each atom (which needs the potential), its number of unlike neighbours, and the size of its cell. The finder needs no potential. For vacancies, interstitials and antisites it finds 1.7–2.4 times as many as the atom energy (Section 5.2).
- Once the defect is found, ordinary energy minimisation does the repair. A network trained to undo the defect learns to move the right atoms, but adds nothing after minimisation and damages the structure without it (Sections 5.3 and 5.5).
- A quarter of our Frenkel pairs undid themselves while the glass relaxed, and left no defect. We found this while checking the physics for this report, removed them from the test, and report the consequences openly (Sections 3.2 and 6.1).
- The noise-removal training is what makes the finder learn quickly. In short training runs that change one setting at a time, a finder that starts from the denoiser finds 44.7% of the defects, and one that starts from random weights 23.7% (Section 5.4).

2 Background

Denoising diffusion models of glasses. Our starting point is DM2 [20]. A network is shown glasses whose atoms have been shifted at random, and learns to predict the shifts so that they can be undone.

Applied again and again with smaller and smaller shifts, it turns random positions into a realistic glass. Hsu et al. [8] used the same idea on crystals for a purpose close to ours: a denoiser removes the jitter of thermal motion, so that the crystal structure underneath, and its defects, can be identified. Diffusion models have also been used to relax crystals that contain given point defects [5]. We are not aware of work that finds and repairs point defects in a glass.

Learned structural indicators. Cubuk et al. [4] trained a classifier on the surroundings of each particle to pick out the particles of a glass that are about to move. Our finder is similar in spirit: a label for each atom, learned from structure alone. But its labels are exact, because we make each defect ourselves.

Resting structures. Letting the atoms settle takes any arrangement to the nearest valley of the energy, called its inherent structure [16]. A glass has an enormous number of such valleys close to each other. All our glasses, with and without defects, rest in a valley, and "the same glass" means the same valley.

3 Methods

3.1 The glass

We use a standard model glass, the two-dimensional Kob–Andersen mixture [3, 11]. It has two species: 65% large A atoms and 35% small B atoms. Two atoms of species α and β at distance r have the Lennard-Jones energy

$$u_{\alpha\beta}(r) = 4\varepsilon_{\alpha\beta}\left[\left(\frac{\sigma_{\alpha\beta}}{r}\right)^{12} - \left(\frac{\sigma_{\alpha\beta}}{r}\right)^{6}\right]. \tag{1}$$

In words: two atoms repel strongly when they come closer than about $\sigma_{\alpha\beta}$, attract each other a little further out, and do not feel each other when far apart. $\sigma_{\alpha\beta}$ sets the size of the atoms and $\varepsilon_{\alpha\beta}$ the strength of the attraction. The values are $\varepsilon_{AA} = 1$, $\varepsilon_{AB} = 1.5$, $\varepsilon_{BB} = 0.5$, $\sigma_{AA} = 1$, $\sigma_{AB} = 0.8$ and $\sigma_{BB} = 0.88$, and the energy is set to zero beyond $2.5\,\sigma_{\alpha\beta}$. We measure lengths in units of $\sigma \equiv \sigma_{AA}$, about one atom diameter, and energies in units of $\varepsilon \equiv \varepsilon_{AA}$. Unlike atoms attract each other most strongly (ε_{AB} is the largest). This, and the 65:35 mixture, keep the atoms from ordering into a crystal [3]. We chose this model because one glass takes seconds to make, so we can afford thousands of defects with exact answers, and because every atom can be drawn and inspected.

Each glass has $N = 256$ atoms in a periodic square box of side 14.6σ. With the simulation program LAMMPS [17] we make it the way real glass is made. We heat the atoms until they form a liquid (temperature $T = 2.0$ in the units of the model), cool the liquid quickly (to $T = 0.01$ at a rate of 10^{-2} per time unit) so that the atoms freeze in disorder, and then let them settle by energy minimisation. The dataset has 800 training, 100 validation and 100 test glasses. Because the cooling is fast, these glasses are not the most stable ones possible, which matters for the antisite (Section 3.2).

3.2 The four point defects

Each defect is made by a known operation on a defect-free glass (Table 2, Figure 1). The names come from crystals. A glass has no lattice site that could be empty or doubly occupied, so here a defect is defined by the operation that made it. After the operation the glass is relaxed: all atoms are free to shift until the forces balance, with the box fixed. The defected glass then rests in a valley of the energy, as a real sample would. Because we know the operation, we know the original glass. It is the exact target for repair, and we need no thresholds to decide what counts as a defect. Each training glass receives 8 defects of each type at random places (about 6,400 examples per type); each validation and test glass receives 2 (200 per type). Glasses of the three sets are never mixed.

What the defects become. We measured each relaxed defect on the test glasses (Table 3).

Vacancies partly close. Right after an atom is removed, the emptiest point of the hole is 0.87σ from the nearest atom. After relaxation this distance has fallen to 0.74σ (median), because neighbours move in. For comparison, around an atom of defect-free glass it is 0.63σ, and only one place in

Table 2: The four point defects and how each is made. The glass is relaxed after the operation.

Defect	Operation on the defect-free glass	Atoms
Vacancy	remove one atom (A or B, chosen at random)	255
Interstitial	add one small B atom at the most open point within 2σ of a random position	257
Frenkel pair	move one atom to the most open point between 1.2σ and 3σ from its place	256
Antisite	exchange the species of an A atom and its nearest B atom	256

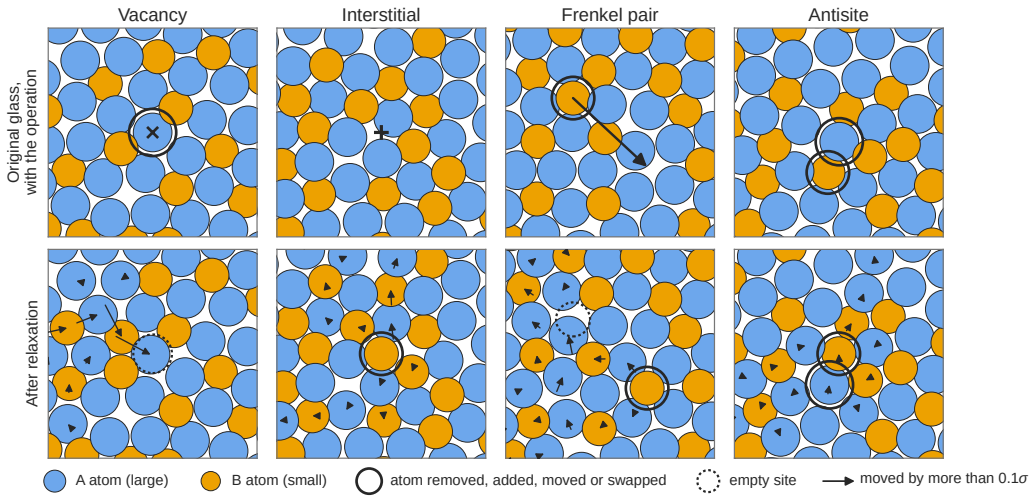


Figure 1: The four point defects in test glasses. Top: the original glass and the operation ($\times$ atom removed, $+$ atom added, arrow: atom moved, two rings: species exchanged). Bottom: the same region after relaxation. Each example is the one with the median number of displaced atoms for its type. In the vacancy example, neighbours have moved into the empty place.

twenty is more open than 0.70σ. About 60% of the relaxed vacancies are still more open than that. The other 40% have handed their empty space to the surroundings and no longer stand out as a hole (as in the example of Figure 1). This agrees with early simulations of model glasses [1, 6].

An interstitial makes room for itself. After relaxation the neighbours have moved apart to take the extra B atom in. It is 0.85σ from its nearest neighbour, only slightly closer than the 0.87σ that is normal for a B atom, and somewhat higher in energy (-3.0 against $-3.4\,\varepsilon$).

A quarter of the Frenkel pairs undo themselves. In 50 of the 200 test Frenkel pairs (and 27% of the training pairs), the moved atom pushed a neighbour of its own species into the empty place and took that neighbour's place. The relaxed glass is then the original glass with two identical atoms exchanged: the same structure with the same energy, and no defect. That a displaced atom and an empty place close to each other cancel out, also by one atom replacing another, is well known from radiation damage in crystals [19]. Our program only checked whether the moved atom had returned to *its own* place, so it missed these cases. We exclude them from the test, which leaves 150 Frenkel pairs; the consequences for training are discussed in Section 6.1.

An antisite is a fault in who sits next to whom, not in how much room there is. After relaxation the neighbours have adjusted their distances to the two exchanged atoms, and the cell of each is close to normal for its species: the cell of the A atom in a B place is slightly large, and that of the B atom in an A place is normal. What remains is the wrong chemistry. Both atoms have lost neighbours of the other species: the A atom has 1.8 unlike neighbours on average where 2.4 is usual, and the B atom 3.8 where 4.4 is usual. Unlike atoms attract each other most strongly in this mixture, so this is also where energy is lost.

An antisite does not always cost energy. The Frenkel pair and the antisite keep all atoms, so their energy can be compared directly with that of the original glass. It is higher for 91% of the Frenkel pairs (median $3.2\,\varepsilon$) and for 89% of the antisites (median $2.8\,\varepsilon$). In the other 11% the swapped and relaxed glass is *lower* in energy than the original. In a stable ordered crystal a swap always costs

Table 3: The relaxed defects on the test glasses. Values are medians (half of the cases lie below, half above), with the middle half of the cases in brackets. "Atoms displaced" counts other atoms moved by more than 0.1σ.

	Vacancy	Interstitial	Frenkel pair	Antisite
Real defects in the test	200	200	150	200
Atoms displaced	14 [3, 43]	22 [12, 40]	44 [21, 69]	29 [15, 55]
Largest displacement of another atom (σ)	0.33	0.32	0.64	0.42
Energy above the original (ε)	–	–	3.2 [1.5, 5.0]	2.8 [1.4, 4.3]

Table 4: The tool used at each stage and the reason for it.

Stage	Tool	Why
Make glasses	simulation of the atoms' motion (molecular dynamics, LAMMPS)	The physical ground truth.
Relax defects; final step of repair	energy minimisation (FIRE [2])	Defines a relaxed glass, and decides which valley a corrected glass falls into.
Learn the glass without labels	graph neural network (EGNN), denoising loss	Learns what normal glass looks like from defect-free glasses only (Section 3.5).
Find the defect	the same network with a label for every atom	The answer must not change when the glass is shifted, turned or its atoms renumbered, and must respect the periodic box.
Repair without a potential (tested)	the same network, denoising loss with a defect as the disturbance	Tests whether the network can replace minimisation.
Baselines for finding	cell size; number of unlike neighbours; atom energy	The simplest measures of empty space and of chemical order, and the standard measure that uses the potential.

energy. A quickly cooled glass is not in its most stable state, and exchanging two atoms can help it to a more stable one; a well-known simulation method, swap Monte Carlo, uses exactly such exchanges to make very stable glasses [13]. The original glass is still the well-defined target for repair, but in these cases it is not the more stable structure.

3.3 Which tool is used at each stage, and why

Table 4 lists every tool in the study. The learned parts are one neural network, used in three ways.

3.4 Model description

A neural network for atoms has to respect three facts. The atoms have no natural order. The glass can be shifted or turned without changing it. And the simulated box repeats in every direction. A *graph neural network* handles all three. It treats the glass as a network in which every atom is a node and atoms close to each other are linked. Each atom carries a list of 128 numbers, its *features*, that describes its surroundings. In every layer each atom receives a message from each of its neighbours and updates its features, so that after several layers the features describe a wider and wider neighbourhood. We use the EGNN design [14], adapted to a periodic box, with two outputs, called heads (Figure 2, Table 5). The layers are as follows.

Input layer. The input is one glass: for each of its N atoms a position $x_i \in \mathbb{R}^2$ in the periodic box and a species $t_i \in \{\mathrm{A}, \mathrm{B}\}$. The denoiser and the repair network also receive the noise level σ_n (how far the atoms were shifted), encoded as 64 random Fourier features of $\log \sigma_n$. The starting features of

Table 5: The deep learning models. $N \times \{\text{layer}\}$ is a layer repeated N times. Values in brackets were tried in the ablations (Section 5.4). Fixed: 128 features, cutoff 2.5σ, 16 radial basis functions, AdamW, batches of 16 glasses.

Model	Layers	Training data, loss and settings	Size
Denoiser	$1\times$\{input embedding\}, $4\times$\{EGNN layer\}, vector head	defect-free glasses with noise, $\sigma_n \sim \mathscr{U}(0.001, 1)$; squared error; 40,000 updates; no labels	0.531 M
Finder	$1\times$\{input embedding\}, $4\times$\{EGNN layer\} (1, 2), role head {Dense 128, Dense 5}	relaxed defected glasses; cross-entropy with class weights (none); start: denoiser (random); 20,000 updates, then 40,000 at half the learning rate (2,000); Frenkel pairs: all (real only)	0.523 M
Repair network, version 1	as the denoiser	corrected defected glasses with noise, $\sigma_n \sim \mathscr{U}(0.001, 0.5)$; squared error; start: denoiser; 20,000 updates	0.531 M
Repair network, version 2	as the denoiser	the same with the displaced atoms weighted 11 times and σ_n log-uniform in $(0.001, 0.3)$; start: version 1; 20,000 updates	0.531 M

atom i are

$$h_i^0 = \mathrm{MLP}\big(E(t_i) + \mathrm{MLP}(\mathrm{Fourier}(\log \sigma_n))\big) \in \mathbb{R}^{128}, \tag{2}$$

where E is a learned look-up table and MLP stands for a small fully connected network; the finder has no noise term. Positions enter only through the graph.

Graph layer. Atoms closer than $r_c = 2.5\sigma$ are linked (about 22 neighbours each), and r_{ij} is the shortest vector from atom i to atom j across the periodic box. Distances are expanded in 16 Gaussian radial basis functions (rbf) and multiplied by a smooth cutoff $f_c(r) = \frac{1}{2}[\cos(\pi r/r_c) + 1]$, so that a neighbour fades out gradually as it reaches the edge of the range.

EGNN layers. Four layers (Figure 2b) pass messages m_{ij} between linked atoms, update the features and produce one vector per atom:

$$\begin{aligned} m_{ij} &= \phi_e\big(h_i, h_j, \mathrm{rbf}(|r_{ij}|)\big)\, f_c(|r_{ij}|), \qquad h_i \leftarrow h_i + \phi_h\Big(\mathrm{LN}(h_i),\ \tfrac{1}{K_0}\textstyle\sum_j g(m_{ij})\, m_{ij}\Big), \\ v_i^\ell &= \tfrac{1}{K_0}\textstyle\sum_j r_{ij}\, \phi_x(m_{ij})\, f_c(|r_{ij}|). \end{aligned} \tag{3}$$

ϕ_e ($272 \to 128 \to 128$), ϕ_h ($256 \to 128 \to 128$) and ϕ_x ($128 \to 128 \to 1$) are two-layer networks with SiLU activations, g is a sigmoid gate, LN is layer normalisation and $K_0 = 24$. Each layer has 117,890 parameters.

Output heads. The *vector head* adds the vectors of the four layers, $d_i = \sum_\ell v_i^\ell \in \mathbb{R}^2$ (no activation). It is an arrow for every atom: the predicted shift of that atom. The denoiser and the repair network use it. The *role head* is a two-layer network ($128 \to 128 \to 5$) on the final features h_i^4. It gives each atom a score for five roles: *intact*, *next to an empty site*, *extra atom*, *displaced atom* and *wrong species*. The finder uses it.

What the equations model. The messages depend only on distances and species. So the features h_i, and with them the role scores, do not change when the glass is shifted, turned or its atoms renumbered. The vector v_i^ℓ is a sum of the arrows to the neighbours, each with a learned weight, so it turns when the glass is turned. This is exactly the form of a force between pairs of atoms, $F_i = \sum_j r_{ij}\, f(|r_{ij}|)$, except that the weight depends on the whole neighbourhood and not on one distance. The vector head is therefore a learned force field. Each layer reaches 2.5σ, so after four layers an atom's output depends on the structure up to 10σ away, about ten atom diameters.

3.5 Loss functions and training

The denoising loss. Following DM2, we take a defect-free glass x_0, shift every atom by random (Gaussian) noise of size σ_n, $x = x_0 + \sigma_n \varepsilon$, and train the network to predict the shift:

$$\mathscr{L} = \mathbb{E}\,\big\| d_\theta(x, \sigma_n) - (x - x_0) \big\|^2. \tag{4}$$

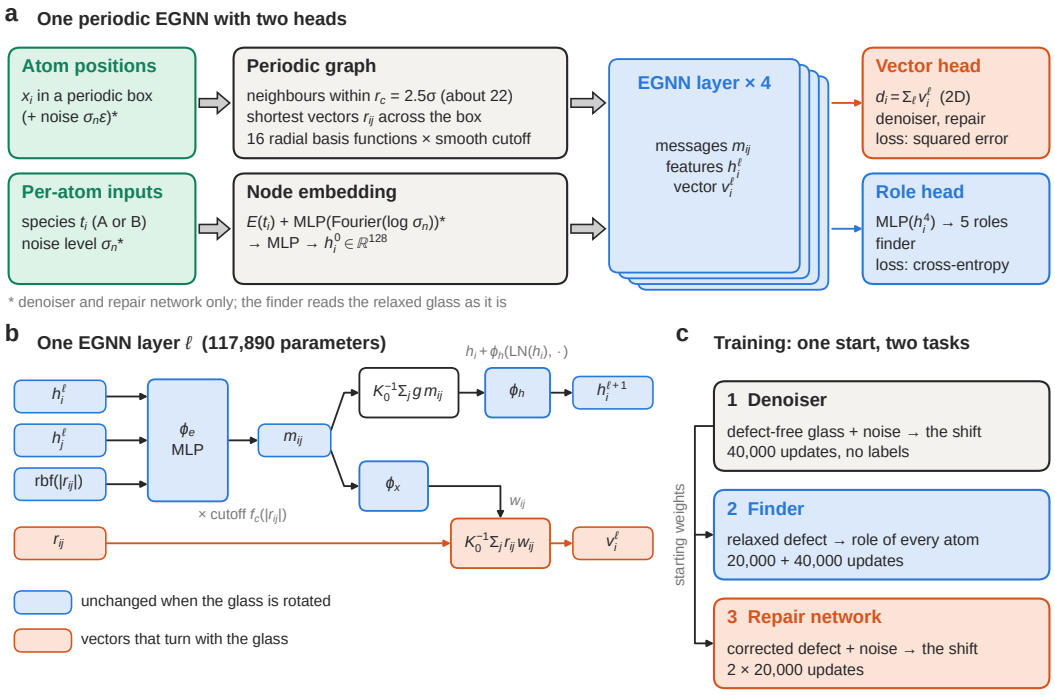


Figure 2: The deep learning model. (a) One network: the positions define a periodic graph, the per-atom inputs define the starting features, and four EGNN layers turn both into features and a vector for every atom. The vector head gives the denoiser and the repair network, the role head the finder. (b) One EGNN layer: blue items do not change when the glass is turned, orange items turn with it. (c) Training: the denoiser is trained first, without labels, and its weights start the finder and the repair network.

Many different glasses could have produced the same noisy structure x, so the best the network can do is to predict the average shift over all of them. A classical result, Tweedie's formula [7, 18], says what this average is: $d^*(x) = -\sigma_n^2 \nabla \log p_{\sigma_n}(x)$, where $p_{\sigma_n}(x)$ says how likely the noisy structure x is. In words: undoing the predicted shift moves the atoms towards more likely structures. For atoms that move freely at a fixed temperature (thermal equilibrium), the likely structures are those of low energy ($\log p = -U/k_BT$ up to a constant, with U the energy and T the temperature), and the direction towards more likely structures is the direction of the physical force. So the loss teaches the network a *restoring force*, although it is never shown a force. Our glasses are not in equilibrium, so we checked this directly: at $\sigma_n = 0.05$ the learned correction and the true force of the model point the same way (average cosine 0.81, where 1 means the same direction), more closely than the straight line back to the original does (0.68).

The finder's loss. The finder is trained on the relaxed defected glasses. The true role y_i of every atom is known from the operation that made the defect. The loss is the cross-entropy, the standard loss for classification, between the role scores and the true roles:

$$\mathscr{L}_{\text{find}} = -\,\mathbb{E}\left[\frac{\sum_i w_{y_i} \log p_\theta(y_i \mid x)_i}{\sum_i w_{y_i}}\right], \qquad w_c \propto (\text{frequency of role } c)^{-1/2}. \tag{5}$$

More than 98% of the atoms are intact, so a network could score well by calling every atom intact. To prevent this, mistakes on the rare roles count more: $w = 1$, 9.2, 31.7, 31.8 and 22.4 for the five roles. Such weights make the network overstate the probabilities of the rare roles. At test time we therefore subtract $\log w_c$ from each score, the standard correction (logit adjustment [12]); it needs no retraining. We call this step *calibration*.

The repair loss. To test repair without a potential, we train the denoiser on one more kind of disturbance. The input is a defected glass in which the number and species of the atoms have already

been corrected, plus noise, $x = x_{\text{corrected}} + \sigma_n \varepsilon$. The target is again the shift to the original glass:

$$\mathscr{L}_{\text{repair}} = \mathbb{E}\left[\frac{\sum_i \omega_i \left\| d_\theta(x, \sigma_n)_i - (x - x_0)_i \right\|^2}{\sum_i \omega_i}\right]. \tag{6}$$

In version 1 every atom counts the same ($\omega_i = 1$). In version 2 the atoms that the defect displaced by more than 0.1σ count 11 times, and most examples have little noise.

Training. The glasses are split 800/100/100 into training, validation and test. Validation glasses are used to choose settings and to watch for over-fitting, test glasses only for the results. The denoiser is trained first, on defect-free glasses only. The finder and the repair network start from its weights (Figure 2c). We use PyTorch and the AdamW optimiser with learning rate 2×10^{-4}. One *update* is one learning step on a batch of 16 glasses. We keep an exponential moving average of the weights (0.999) and use it for the tests. One batch in five contains defect-free glasses only, so that the networks learn to leave intact glass alone. Each run takes between one and two and a half hours on one T4 GPU.

3.6 Evaluation metrics

All tests use the same 750 real defects from the 100 test glasses, and every rate is a fraction of all defects of that type.

Found answers the question "is the defect where the method points?". Each method (the finder, and the baselines of Section 4) gives every atom a score and flags the atom with the highest score; for the finder the score is the probability of the role that belongs to the defect. The defect is *found* if the flagged atom lies within 1.5σ of it (of either end, for a Frenkel pair or an antisite); 1.5σ covers the nearest neighbours. A randomly chosen atom is that close only 3–5% of the time.

Restored answers the question "did we get the original glass back?". The repair has two steps (Figure 3). First the number or species of atoms is corrected at the flagged atom: for a vacancy, an atom of the missing species is put at the most open point within 1.5σ; for an interstitial, the flagged atom is removed; for a Frenkel pair, the flagged atom is moved to one of up to eight open points 1–3.2σ away; for an antisite, a neighbouring A–B pair near the flagged atom is exchanged. Where there is a choice (which open point, which pair), each method chooses with its own score, and the energy route tries every option and keeps the one with the lowest energy. Then the energy is minimised with all atoms free. The glass is *restored* if the result is the original glass: every original atom has an atom of its species within 0.05σ, one to one. Identical atoms may have exchanged places, since that does not change the structure.

The type of defect is taken as known in these tests; only its location is not. As an upper limit we also run the repair with the true location given.

Uncertainty. With 150–200 defects of each type, a rate is only known to within several points. The brackets in our tables give the range that holds the true rate with 95% confidence. We obtain it by drawing the 100 test glasses again at random, 10,000 times (a bootstrap); whole glasses are drawn because the two defects made in one glass are not independent. To compare two methods we look at the same defects and count those that only one of the two gets right (McNemar's test). The resulting p is the probability that a lead of this size arises by chance; below 0.05 we call a difference significant.

3.7 Experimental iterations

The main results took five training runs (the denoiser, the finder in two stages and two versions of the repair network; Table 5), and the ablations six short ones. Each step answered one question; Table 6 gives the numbers.

1. *Is training on defects needed?* The denoiser alone, which has never seen a defect, already finds 44% of the vacancies by the size of its correction, but few interstitials and antisites. So we trained the finder on labelled defects, starting from the denoiser.

Table 6: The comparisons made on the way, on the test defects (%). The numbers at the left are those of the experimental iterations. †Measured with the finder after 20,000 updates.

	Experiment	Compared	Measure	Result
1	Training on defects	denoiser alone → trained finder	found: vacancy / interstitial / antisite	44 / 12 / 15 → 69.5 / 50 / 49
2	Training length	20,000 → 60,000 updates	found, all 750 defects	50.3 → 54 ($p = 0.001$)
3	Calibration of the finder	off → on	defect-free glasses flagged; found, all defects	68 → 2; 50.7 → 54 ($p < 0.001$)
4	Loss of the repair network	version 1 → version 2	error on the displaced atoms removed by one step	7–9 → 13–16
	Steps of the repair network	1 → 10 steps (version 2)	glasses left with overlapping atoms	1–40 → 50–73
	Repair network before minimisation	without → with (version 2)	restored: vacancy / interstitial†	51 / 34.5 → 51.5 / 34
5	Check of the labels	all 200 → the 150 real Frenkel pairs	Frenkel pairs found / restored	37 / 18 → 45.3 / 12.7

2. *Is the finder trained long enough?* Continuing for 40,000 updates at half the learning rate found a few more defects. Over the last 20,000 updates the validation scores moved by two points or less, so we stopped.
3. *Do its probabilities mean anything?* The class weights made the finder see defects in most defect-free glasses. Calibration corrected this without retraining.
4. *Why did the first repair network not learn the defect?* A defect displaces a few tens of the 256 atoms, so the random noise on all atoms filled the loss. Version 2 weights the displaced atoms up and trains mostly at small noise.
5. *Are the labels right?* The physics check found the Frenkel pairs that had undone themselves (Section 3.2). All tests were repeated without them.
6. *Which design choices matter?* Six short runs change one setting at a time: the starting weights, the number of layers, the class weights and the cleaned training set (Section 5.4).

4 Baselines and extensions

Baselines (implemented and run by us). Three measures that involve no learning flag one atom per glass, as the finder does, and are followed by the same repair steps and the same minimiser:

- *Cell size* (no potential, no learning): the area of each atom's cell, compared with what is usual for its species in defect-free glasses. It measures how much room an atom has [15]. We flag the largest cell for a vacancy, the smallest for an interstitial, and the most unusual for a Frenkel pair or an antisite.
- *Unlike neighbours* (no potential, no learning): the number of nearest neighbours of the other species, compared with what is usual in the same way. It measures chemical order, that is, who sits next to whom. Each defect leaves atoms with too few unlike neighbours for their species (the neighbours of an empty place, a squeezed-in extra atom, the two exchanged atoms), so we flag the atom with the fewest.
- *Atom energy* (needs the potential): each atom's share of the total energy, compared with what is usual in the same way. The highest is flagged. Forces cannot be used, because in a relaxed glass they are all zero.

The rules for the two structural measures (largest, smallest, fewest) were chosen as the best on the validation glasses (Appendix A). Two reference values frame the results: a randomly chosen atom, and the repair with the true location given.

Reproduction evidence. Every number comes from our own runs on the same 750 defects. The atom-energy baseline was computed by two separate programs, which agree on every single defect. The evaluation of the finder, repeated on a different machine (CPU in place of GPU), gave the same finding rates. The denoiser is our implementation of the DM2 training [20] for the two-dimensional glass.

Extensions beyond DM2. DM2 is a denoiser used to generate glasses. We add: (i) the role

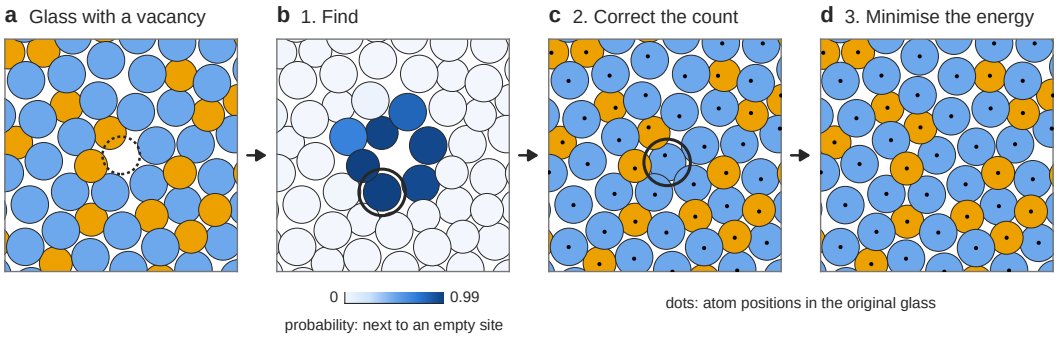


Figure 3: Finding and repairing one test vacancy. (a) The relaxed glass; the dotted circle marks where the atom was removed, which the methods are not told. (b) The finder's probability that each atom is next to an empty site; the flagged atom is ringed. (c) An atom of the missing species is put at the most open point near the flagged atom. (d) After energy minimisation every atom sits on its position in the original glass (dots); in (c) the atoms around the defect are still off theirs.

Table 7: Defects found (%): the flagged atom lies within 1.5σ of the defect. Brackets: the range that holds the true rate with 95% confidence (Section 3.6). Only the atom energy uses the interatomic potential. Bold: the best method.

		No potential			Potential	
Defect	n	Learned finder	Cell size	Unlike neighbours	Atom energy	Random atom
Vacancy	200	**69.5** [62, 76]	47.5 [40, 55]	29 [22, 36]	29.5 [23, 36]	2.9
Interstitial	200	**50** [42, 57]	15 [10, 20]	28 [22, 34]	28 [22, 34]	3
Antisite	200	**49** [42, 56]	16.5 [12, 22]	31.5 [25, 38]	29 [23, 35]	4
Frenkel pair	150	**45.3** [38, 53]	26.7 [20, 33]	27.3 [21, 34]	32.7 [25, 40]	5.4

head and its class-weighted loss, which turn the denoiser into a finder; (ii) calibration of the finder's probabilities; (iii) a defect, and not only noise, as the disturbance to remove, with the displaced atoms weighted up; (iv) tests against measures without learning, on the same defects, with paired statistics.

5 Results

5.1 Before any training on defects

The denoiser, trained only on defect-free glasses, already carries information about defects. Ranking the atoms by the size of the correction it predicts finds 44% of vacancies, against 29.5% for the atom energy. It is poor for interstitials and antisites (12% and 15%), and it cannot say which species belongs where. This is the reason for training on defects, and for starting that training from the denoiser.

5.2 Finding the defect

Table 7 and Figure 4a give the main result. The learned finder locates 69.5% of vacancies, 50% of interstitials and 49% of antisites. The best of the three other measures finds 47.5%, 28% and 31.5%, so the finder is ahead by 18–22 points. Compared on the same defects, its lead over each measure is significant for each of these three types ($p < 0.001$). The finder also locates 45.3% of the 150 real Frenkel pairs, against 32.7% for the atom energy ($p = 0.014$) and less for the two structural measures. A Frenkel pair has two ends, the empty place and the displaced atom, and a flag near either end counts as found. The finder flags exactly the displaced atom, which the repair has to move back, in only 21% of the pairs (the atom energy in 7%).

The three other measures fail in ways that say something about the defects. *Cell size* is a serious competitor only for vacancies (47.5%), where missing volume is what one looks for. It finds few interstitials and antisites (15% and 16.5%), because after relaxation their cells are hardly unusual

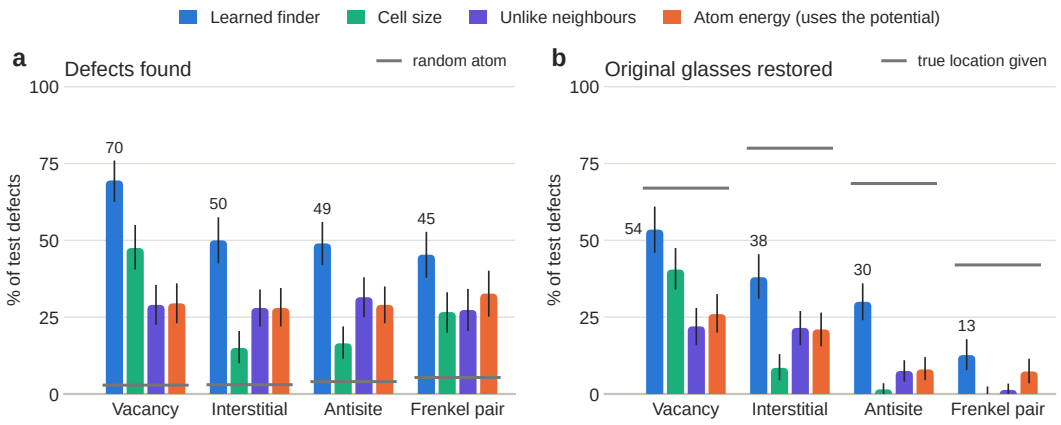


Figure 4: (a) Defects found: the flagged atom is within 1.5σ of the defect. Grey lines: a randomly chosen atom. (b) Original glasses restored after correcting the atoms at the flagged atom and minimising the energy. Grey lines: the same with the true location given. Whiskers: the range that holds the true rate with 95% confidence. The first three methods use no potential to flag the atom. The numbers are the learned finder's rates; all values are in Tables 7 and 8.

Table 8: Original glasses restored (%) after the atoms are corrected at the flagged atom and the energy is minimised (every route uses the potential for this last step). Columns: how the atom was flagged. Last column: the true location given. Brackets as in Table 7.

Defect	Learned finder	Cell size	Unlike neighbours	Atom energy	Location given
Vacancy	**53.5** [46, 61]	40.5 [34, 48]	22 [16, 28]	26 [20, 32]	67 [60, 74]
Interstitial	**38** [31, 46]	8.5 [4, 13]	21.5 [16, 27]	21 [16, 26]	80 [74, 86]
Antisite	**30** [24, 36]	1.5 [0, 4]	7.5 [4, 11]	8 [4, 12]	68.5 [62, 74]
Frenkel pair	**12.7** [8, 18]	0 [0, 2]	1.3 [0, 3]	7.3 [3, 11]	42 [34, 50]

(Section 3.2). *The number of unlike neighbours* finds 27.3–31.5% of every type, close to what *the atom energy* finds (28–32.7%). This is no coincidence. In defect-free glass an atom's energy follows its number of unlike neighbours closely (correlation -0.90 for A atoms and -0.84 for B atoms, where -1 would be a perfect straight-line relation), because unlike atoms attract each other by far the most strongly. For finding defects, the potential therefore adds little to a simple count of neighbours.

All three are limited by the same thing. A relaxed defect has spread its energy and its empty space over many atoms (Table 3), and a glass has many atoms that are naturally loosely packed, short of unlike neighbours or high in energy. The atom with the highest energy is at the defect about a third of the time or less.

5.3 Restoring the original glass

Table 8 and Figure 4b show what happens when the correction and the minimisation follow. With the learned finder, 53.5% of vacancy glasses, 38% of interstitial glasses and 30% of antisite glasses return to their original structure. The best of the other routes restores 40.5%, 21.5% and 8% ($p < 0.001$ for the finder against each route). Frenkel pairs are the hardest to restore: 12.7% with the finder, 7.3% with the energy route, which tries every open point and keeps the lowest energy, and 0% and 1.3% with the two structural measures. Two things have to be right here, the displaced atom and the open point it came from, and the finder's lead over the energy route could be chance ($p = 0.169$).

Two things stand out. First, restoring follows finding. With the finder, the restored rate is about three quarters of the found rate for vacancies and interstitials. It is lower for antisites and much lower for Frenkel pairs, where a second choice has to be right as well: which pair to exchange, or which open point to move the atom to. Second, the upper limit is far above every method. With the true location given, minimisation alone restores 67% of vacancies, 80% of interstitials, 42% of Frenkel pairs and 68.5% of antisites. What is missing is better finding, not better repair. The upper limit is

Table 9: Ablations of the finder: defects found (%) on the real test defects after 2,000 updates, one setting changed per run. Size in millions of parameters. Brackets: the range that holds the true rate with 95% confidence. p: the probability that the difference from the run it is compared with is chance; that run is the reference, or, for the two runs with fewer layers, the run with random starting weights and 4 layers.

Run	Size	Vacancy	Interst.	Antisite	Frenkel	All	p
Reference: starts from the denoiser, 4 layers, class weights	0.52	64	29.5	42.5	42	44.7 [40, 49]	–
Random starting weights	0.52	17	27.5	26.5	24	23.7 [20, 27]	< 0.001
Random starting weights, 2 layers (reach 5σ)	0.29	15.5	27.5	25	24.7	23.1 [20, 27]	0.46
Random starting weights, 1 layer (reach 2.5σ)	0.17	13.5	23	15.5	17.3	17.3 [14, 20]	< 0.001
No class weights	0.52	61	31.5	41	36.7	42.9 [39, 47]	0.12
Trained without the Frenkel pairs that undid themselves	0.52	64	29.5	38	35.3	42.1 [38, 46]	0.027

below 100% because a glass has many valleys of the energy close to each other: relaxing the defect moved the neighbours, and putting the atom back does not always lead down into the valley the glass started in.

5.4 Ablations of the finder

An ablation changes or removes one part of a method to see how much that part matters. Table 9 does this for the finder, one setting per run. Six full training runs were beyond our computing budget, so each run has only 2,000 updates, where the finder of the main results has 60,000. The rates are therefore lower than in Table 7, and the runs should be compared with each other only. There is one run per setting, so a difference of two or three points can also come from the randomness of training. The short runs are scored with their final weights, not with the moving average used for the long runs: an average over so few updates still contains 13% of the starting weights, which would count against a run that starts from random weights.

Starting from the denoiser matters most. After the same number of updates, a finder that starts from random weights finds 23.7% of the defects, and the reference, which starts from the denoiser, 44.7% ($p < 0.001$). The denoiser was trained on defect-free glasses without any labels and already knows what normal glass looks like; the finder then only has to learn what a defect changes. The gain is largest for vacancies (17% against 64%), the defect that the denoiser alone already found best (Section 3.7). There is none for interstitials (27.5% against 29.5%), where the denoiser alone was poor. These short runs cannot say whether a network that starts from random weights would catch up with much longer training.

The finder has to see beyond the nearest neighbours. Each layer adds 2.5σ to the distance over which the label of an atom can depend on the structure. From random weights, one layer finds 17.3% of the defects, two layers 23.1% and four layers 23.7%. One layer is clearly worse ($p < 0.001$); two and four do equally well at this budget ($p = 0.46$). This agrees with the picture of a relaxed defect as a pattern over a few tens of atoms (Table 3). Whether the full reach of four layers pays off cannot be seen in runs this short.

Class weights change the labels, not the ranking. Without the class weights the finder finds 42.9% of the defects, and with them 44.7%; the difference could be chance ($p = 0.12$). "Found" only asks which atom is ranked first, and the weights do not change that. They change how readily the network gives an atom a defect label. Trained without them, the network plays safe and calls nearly every atom intact, which is right 98% of the time: on validation glasses it labels 7% of the atoms next to a vacancy as such, and no extra atom at all. With the weights these rates are 27% and 18%.

Cleaning the training set did not help. We expected the finder to improve when the Frenkel pairs that had undone themselves are taken out of the training set, since they carry defect labels but hold no defect. In this short run it did not: 35.3% of the Frenkel pairs are found against 42%, and 42.1% against 44.7% of all defects ($p = 0.027$). We do not read this as a real loss. The p value compares these two trained networks only; it says nothing about how much a result changes when the same training is simply repeated. Scored with the moving average of their weights, the two runs are

equal (44.5% and 44.8%). What the run does show is that taking out the wrong labels brings no gain, at least early in training.

5.5 The repair network

We expected the repair network to be the centre of this study. It was not. Its first version learned to remove noise but not the defect, because the defect moves only a few tens of atoms and the random noise on all 256 filled the loss. The second version, with the displaced atoms weighted up, did learn the defect: on the test defects, one step removes 13–16% of the error on the displaced atoms, against 7–9% for the first version.

It still does not help. When minimisation follows, applying the repair network first changes the restored rates by less than their uncertainty (vacancies 51.5% with it and 51% without; interstitials 34% and 34.5%, measured with the finder after 20,000 updates). Once the right atom is added or removed, minimisation finishes the repair by itself. Without minimisation, the network moves the displaced atoms closer to their original positions, for vacancies and Frenkel pairs slightly closer than minimisation does. But it pushes atoms onto each other: 20% of the vacancy glasses and 40% of the Frenkel glasses end with two atoms closer than is ever seen in the data. Applying it repeatedly makes this worse. A network trained only on distances to a target does not know that atoms cannot overlap. The interatomic potential supplies exactly that knowledge.

5.6 Is there a defect at all?

The finder ranks the atoms of a glass that is known to contain a defect. Deciding whether a glass contains one at all is a different question. Without calibration, the finder gives some atom a defect probability above one half in 68% of *defect-free* glasses. With calibration this falls to 2%, but then only 28.7% of defected glasses are flagged. Ranking glasses by their highest defect probability separates defected from defect-free ones with a score of 0.75 (the area under the ROC curve: 0.5 is chance, 1 is perfect). The finder is useful for saying where a defect is, and not yet for saying whether there is one. Calibration also helps finding: over all test defects the finder locates 54% with it and 50.7% without ($p < 0.001$), most of the gain coming from Frenkel pairs.

6 Discussion

Why a learned model. The question from our review was why artificial intelligence should be used at all, when defects can be made without it. Making defects needs no learning, and we no longer use a model for it. Finding them is different. The finder was shown examples of relaxed defects and then locates new ones from the positions and species of the atoms alone, more reliably than two simple structural measures and than the energies from the very potential that made the glass. It can therefore be applied where no potential is available. The simulation's role is to supply examples and to check the answers.

What the network adds, physically. The denoising loss gives the network a model of the forces in normal glass (Section 3.5), and the finder is built on those features. Each simple measure sees one side of a defect, empty space or chemical order, on one atom at a time. A relaxed defect is not marked by one atom with too much room or too few unlike neighbours, but by a pattern over many atoms (Table 3). A network that passes messages four times, so that each atom's label depends on the structure up to ten atom diameters away, can read such a pattern, and both sides of it at once.

Repair is a job for physics. Once the right atom is added, removed or exchanged, energy minimisation is fast, exact and free of the overlaps that the repair network produces. We see no reason to replace it when a potential is available. When none is available, a repair network would need at least a built-in rule that atoms cannot overlap.

6.1 Limits

- **The Frenkel labels used in training were partly wrong.** 27% of the training Frenkel pairs had undone themselves (Section 3.2) but were still labelled as defects, so the finder was taught to call some intact atoms displaced. All test numbers here exclude such cases. A short training run without them found no more defects (Section 5.4), so this does not seem to hold the finder back; the finder of the main results has not been retrained without them.
- **The type of defect is given.** The finder is asked for a vacancy, an interstitial, a Frenkel pair or an antisite. A count of the atoms separates the first two from the others, but not a Frenkel pair from an antisite.
- **One defect per glass,** in a small box (256 atoms). Real damage produces many defects that interact.
- **A model glass in two dimensions,** at zero temperature (the atoms do not vibrate). Its atoms are simple spheres without the directed chemical bonds or electric charges of real materials. The defects that matter in a real glass such as silica are of a different kind (broken or extra bonds), which is why we started a study there (Section 7).
- **"Restored" is strict.** It asks for the exact original valley of the energy. In a glass, other valleys close by are equally natural, and for 11% of the antisites the defected glass is the more stable one.
- **Uncertainty.** With 150–200 defects per type from 100 glasses, the rates are known to about ± 7 points.

7 Conclusion and next steps

In a model glass, a graph neural network that starts from a denoising diffusion model and is trained on relaxed defects finds point defects without an interatomic potential, more often than the potential's own atom energies and more often than two simple structural measures. Correcting the atoms where it points and minimising the energy then restores more vacancy, interstitial and antisite glasses than any of the other routes; for Frenkel pairs every route restores few. A network trained to undo the defect directly is not needed. The denoising stage earns its place: in short runs, a finder that starts from the denoiser finds nearly twice as many defects as one that starts from random weights. The room for improvement is in finding: given the true location, minimisation restores 42–80% of the glasses.

Next steps, in order:

1. Let the finder name the type of defect, and repeat the training with several random starts to measure how much the results vary from run to run.
2. Test whether repaired glasses stay stable when the atoms are set in motion (molecular dynamics), and repeat the study with several defects per glass.
3. Silica glass (amorphous SiO_2, the main component of window glass). A first stage without any learned model is complete: with the true location given, minimisation restores 76–96% of four standard silica defects, but the standard method there, counting the neighbours of each atom, finds only 11–29% of them. As in two dimensions, finding is the bottleneck, which justifies training a three-dimensional finder.

Team contributions

Workstream	What it covers	Member
Data and physics	glass generation, potential, relaxation, physical checks of the defects	Nicholaus Ladislaus
Defects and evaluation	defect operations, success tests, baselines, statistics	Stacy Cheruto
Models and training	EGNN denoiser, finder and repair network; training runs	Abubakar Chilala
Repair procedure	correction steps, choice of gap and pair, repair without a potential	Gentille Uwera
Baselines and writing	figures, reports	Michel Sangwa
Supervision	course teaching assistant; supervised the project	Boniface Godwin

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Code and data. Code, datasets, trained models and every result in this report: `https://github.com/chilalaabubakar/IDL-PROJECT` (branch `dabo-point-defects`). The numbers of this report are in `results/point_defects_recheck_20261111/` and `results/point_defects_ablations_20261112/`.

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A Details

Parameters. Denoiser and repair network: 530,696 (input embedding 59,136, of which 24,832 encode the noise level; four EGNN layers of 117,890 each). Finder: 523,021 (input embedding 34,304; the same four layers; role head 17,157). The embedding still holds two inputs of the first version of the project, a flag for held atoms and a defect label; both are fixed to "none" in this study.

Relaxation. Defects are relaxed with the FIRE minimiser of LAMMPS until the largest force is below 10^{-8}, with the box fixed; a vacancy or an interstitial therefore changes the density by 0.4%. The repair tests use a FIRE minimiser written in PyTorch (largest force below 10^{-6}), the same for every method, because it can start from overlapping atoms. The two reach the same minimum from a defect placed in a gap.

Open points. "The most open point" is the point of a grid of spacing 0.05σ that is farthest from every atom, within the stated distance range.

The two structural measures. Nearest neighbours are atoms closer than $1.4\,\sigma_{\alpha\beta}$. In defect-free glass an A atom has 2.4 ± 0.9 unlike neighbours and a B atom 4.4 ± 0.5; cell areas are 0.88 ± 0.07 and $0.75 \pm 0.02\,\sigma^2$. Each atom is compared with the mean and spread for its species. Atoms with the same number of unlike neighbours are ranked at random. On the validation glasses we compared five rules for the cell (largest, smallest, most unusual, and two that depend on the species) and two for the neighbours (fewest, most), and kept the best for each defect; no other rule found more. In the repair of a Frenkel pair, the cell-size route moves the atom with the smallest cell to the open point whose neighbours have the largest cells, and the neighbour route moves the atom with the fewest unlike neighbours to the open point whose neighbours are shortest of them. For an antisite, each exchanges the neighbouring A–B pair that is most unusual by its own measure.

Intervals. Where a rate is exactly 0%, drawing the glasses again gives no spread, and the exact binomial interval is shown instead.

Reproduction. From the repository root:

- the test evaluation on real defects: `scripts/pilots/recheck_real_defects.sh`. It calls `scripts/eval_point_defects.py` (the finder), `scripts/pilots/baseline_routes.py` (the three other measures) and `scripts/eval_point_defect_repair.py` (true location given);
- the experiments on the finder: training with `scripts/pilots/finder_ablations.sh`, tests with `scripts/pilots/finder_ablations.py`;
- Table 3: `scripts/pilots/point_defect_physics.py`;
- the cells and neighbours of the antisite atoms, and the choice of rules for the structural measures: `scripts/pilots/structural_signatures.py`;
- numbers, tables and figures of this report: `paper/make_report_numbers.py` and `paper/make_report_figures.py`.